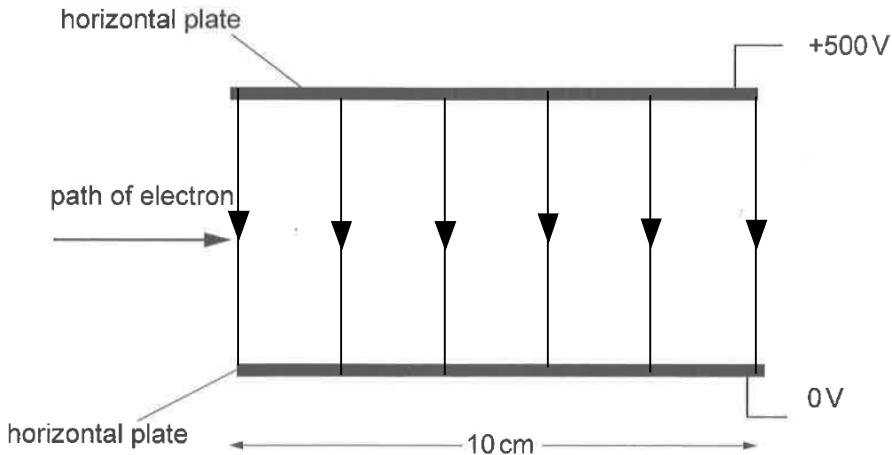
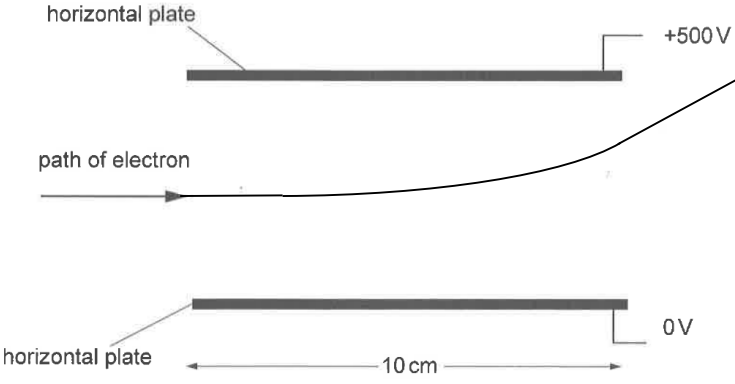


7(a)	Electric field strength at a point is defined as the electric force exerted per unit positive charge placed at that point.
7(b)	Work done by electric field = gain in K.E. of the electron $F_E = qE = (1.6 \times 10^{-19})(3.0 \times 10^5) = 4.8 \times 10^{-14} \text{ N}$ work done = gain in K.E. $F_E d = \frac{1}{2} m_e v^2 - 0$ $(4.8 \times 10^{-14})(1.0 \times 10^{-2}) = \frac{1}{2} (9.11 \times 10^{-31}) v^2 - 0$ $v = 3.25 \times 10^7 \text{ m s}^{-1}$
7(c)(i)	
7(c)(ii)	Force to charge ratio = $\frac{F_E}{e} = \frac{e\left(\frac{V}{d}\right)}{e} = \frac{V}{d} = \frac{500}{5.0 \times 10^{-2}} = 1.0 \times 10^4 \text{ N C}^{-1}$
7(c)(iii)	$F_E = m_e a$ $a = \frac{F_E}{m_e} = \frac{e\left(\frac{V}{d}\right)}{m_e} = \frac{(1.6 \times 10^{-19})(500)}{(9.11 \times 10^{-31})(5.0 \times 10^{-2})} = 1.76 \times 10^{15} \text{ m s}^{-2} = 1.8 \times 10^{15} \text{ m s}^{-2}$ The acceleration is directed upwards.

No.	Solution
7(c)(iv)	In between the parallel plates, the electron moves in a parabolic path. Horizontally, $s_x = u_x t$ $10 \times 10^{-2} = (3.25 \times 10^7) t$ $t = 3.0 \times 10^{-9} \text{ s}$ Vertically, $s_y = u_y t + \frac{1}{2} a_y t^2 = 0 + \frac{1}{2} (1.76 \times 10^{15}) (3.0 \times 10^{-9})^2$ $= 8.3 \times 10^{-3} \text{ m}$
7(d)(i)	 - parabolic path in between the plates. - straight path after exiting the electric field.
7(d)(ii)	The vertically upwards acceleration due to the electric field is $1.8 \times 10^{15} \text{ m s}^{-2}$ and this is very much larger than the downwards acceleration due to gravity of 9.81 m s^{-2}. Thus the path of the electron is not noticeably affected by gravity.
7(e)	$I = \frac{Q}{t} = \frac{N}{t} e$ $\frac{N}{t} = \frac{I}{e} = \frac{0.15}{1.6 \times 10^{-19}} = 9.4 \times 10^{17}$